

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular & Supplementary Summer 2024 Course: B. Tech. Branch : Mechanical Engineering Semester : VI Subject Code & Name: BTMC602_Y22Machine Design-II Max Marks: 60 Date: 15/06/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Solve the of Following MCQ's 1) A Thrust Bearings supports ----- load. (A) Axial (B) Radial (C) Axial & Radial (D) None of the above 2) In Sliding Contact Bearings -----is used to reduce the friction over shaft. (A) Rubber seal (B) Brass bush (C) Balls (D) All of the above 3) Bevel gears are used to connect two shafts which are ----- to each other. (A) Perpendicular (B) parallel (C) Perpendicular but non intersecting. 4) In Industry the shop floor motors are usually provided with ----- (A) Flat Belt drive (B) Gear drives (C) Rope drives (D) V-Belt drives 5) In a helical gears, the helix angle varies from -----degree, (A) 10 to 15 (B) 15 to 20 (C) 15-25 (D) 20 to 25 6) In case of single plate clutch the spring force is used for -----the clutch. (A) Disengaging (B) Engaging (C) Both A & B (D) Neither A nor B.	1/1	6
B)	Write short Notes :- 1) Bearing Failure –Causes and remedies. 2) Rolling contact bearing lubrication.	1/2	6 (2X3)
C)	A ball bearing is operating on a work cycle consisting of three parts – a radial load of 2500 N at 1400 rpm for one quarter cycle , a radial load of 4000 N at 700 rpm for one half cycle and radial load of 3000 N at 1400 rpm for remaining cycle. The expected life of the bearing is 10,000 hr. Calculate the dynamic load carrying capacity of the bearing.	2/3	6
Q.2	Solve Any Two of the following.		12
A)	Define the Following Terms used in gear drives:- 1) Circular pitch, 2) Velocity Ratio , 3) Interference & Undercutting	1/2	6 (2X3)
B)	A pair of spur gears with 20 degree full depth involute teeth consists of pinion having 20 teeth meshing with a gear having 40 teeth. The module is 3 and face width is 36 mm. The material of pinion and gear is steel with a ultimate tensile strength of 500 N/mm ² . The gears are heat treated to surface hardness of 400 BHN .The pinion rotates at 1400 rpm and service factor for the application is 1.50. Assume the velocity factor accounts for the dynamic load and factor of safety is 1.25 . Determine the power that the gear can transmit. Assume Lewis form factor Y =0.32.	2/3	6

C)	A pair of parallel helical gears consists of 26 teeth pinion running at 3000 rpm and transmitting 8 KW power to gear. The speed ratio is 4. The normal pressure angle is 20 degree and helix angle is 45 degree. Both the gears are made of same material having ultimate strength of 500 N/mm ² . The service factor and factor of safety are 1.5 and 2 respectively. Calculate:-1) Normal Module , considering velocity factor accounts for the dynamic load and that the face width is ten times the normal module. The pitch line velocity is 10 m/s. 2) Consider the first preference value of normal module and find out main dimensions of the gears. Assume form factor $Y = 0.438$	2/3	6
Q. 3	Solve Any Two of the following.		12
A)	Write short Notes :- 1) Types of bevel gears, 2) Material selection for worm and worm wheel.	1/2	6 (2X3)
B)	A pair of straight bevel gears mounted on shaft are intersecting at right angle. The number of teeth on pinion and gear are 20 and 26 respectively. The pressure angle is 20 degree. The pinion shaft is connected to an electric motor developing 4 Kw. power at 1400 rpm . The service factor is 1.25 . The pinion and gears are made of steel having ultimate strength of 700 N/mm ² and heat treated to surface hardness of 350 BHN. The manufacturing error between meshing teeth is 10 micro meter. The module and face width are 3 mm and 15 mm respectively. Find out the factor of safety against bending as well as pitting failure. Assume form factor $Y = 0.345$ and $C = 11400 \text{ N/mm}^2$	3/4	6
C)	A pair of worm and worm wheel is designated as 3/60/10/6 the worm is transmitting 4 KW at 1400 rpm to the worm wheel .The coefficient of friction is 0.1 and normal pressure angle is 20 degree. Determine the components of the gear tooth force acting on the worm and worm wheel.	2/4	6
Q.4	Solve Any Two of the following.		12
A)	1) Explain the sleep and creep phenomena applied to belt drives. 2) Illustrate with the help of neat diagram the different methods used to join the flat belts.	1/2	6 (2X3)
B)	A belt drive is used to transmit 100 KW power from pulley 0.9 m to another pulley 1.2 m in diameter separated by a distance of 3.4 meter. The belt speed is 20 m/sec and the coefficient of friction is 0.25. There is 1.2 % sleep and 5% friction loss at each shaft and 20 % over load . Design the drive and find out the dimension of the belt assuming belt stress of 2.5 Mpa & thickness 12 mm. & belt density of 1000 Kg/m ³	2/3	6
C)	An electric motor 80 KW running at 700 rpm runs a compressor at 300 rpm using V belt drive. The diameter of the pulley mounted on compressor shaft is 1 meter and the center distance between two pulley's is 1.5 meter. The belt speed is limited to 1500 m/min. Determine the number of V belts required to transmit the power if each belt has cross sectional area of 350 mm ² , density 1000 kg/m ³ and an allowable tensile stress of 2.5 Mpa. The groove angle of the pulley is 35 degree. The coefficient of friction between belt and pulley is 0.25. Also determine the length of each belt required.	3/5	6

Q. 5	Solve Any Two of the following.		12
A)	1) What is the difference between clutch and brakes. 2) Explain in brief self-energizing and self-locking block brake.	1/2	6 (2X3)
B)	A centrifugal clutch running at 1400 rpm consists of 4 shoes each having a mass of 1.25 kg. The center of gravity of each shoe in its engaged position is 100 mm, while the inner radius of the drum is 130 mm. The coefficient of friction is 0.25. The spring force at the beginning is 600 N. Calculate:- 1) the speed at which engagement begins, and 2) the power transmitted by clutch at 1400 rpm.	2/3	6
C)	A double block brake is as shown in the fig.Q 5 C . The brake drum rotates in clockwise direction and actuating force is 400 N. The coefficient of friction between block and drum is 0.30 . Calculate 1) the torque absorbing capacity of the brake.2) the dimension of block if the intensity of pressure between block and brake drum is limited to 1 N/mm² . Assume that the block are identical and the length of each block is twice its width. Q.5. C. Fig.	2/3	6
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